

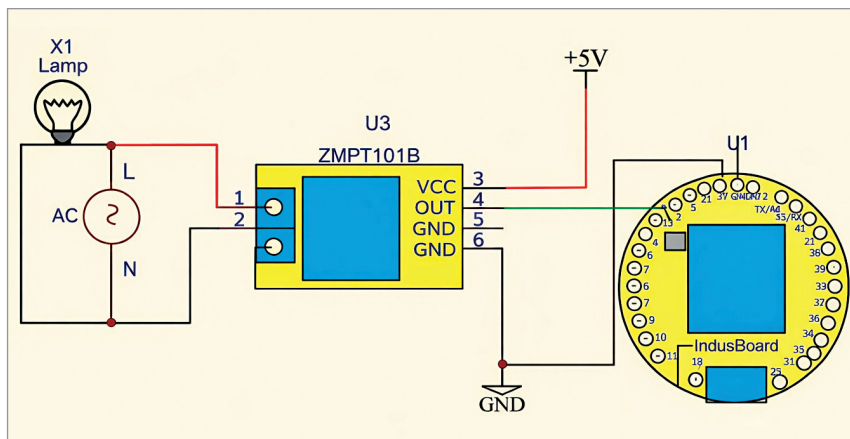


Fig. 3: Calibration circuit

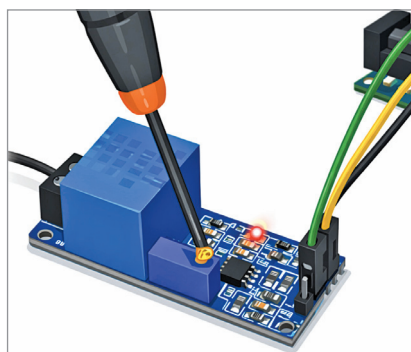


Fig. 4: Rotate the potentiometer clockwise or anticlockwise to obtain the sine waveform

a unified interface for parameter monitoring, remote appliance control, and mesh-network creation to ensure reliable connectivity.

Dedicated current and voltage sensors measure electrical parameters accurately. By combining data from both sensors, real-time energy consumption is calculated. Based on this information, advanced features such as over-voltage and over-current protection, automatic low-power mode, and energy-usage-based automation are implemented.

For example, the system can automatically turn off high-power appliances when the electricity bill reaches a predefined limit, or shut down all connected loads if total consumption exceeds a specified threshold within a billing cycle. Such tightly integrated, energy-aware automation was not feasible in earlier designs.

Conventional solutions either

provide IoT-based control for lights and fans or use separate energy meters for billing, often requiring independent touch switches and control units. In contrast, this design integrates smart switching, energy monitoring, billing intelligence, and automation into a single compact device.

This integrated approach offers flexibility for home automation, real-time monitoring, intelligent switching, and AI-assistant-based voice control. Earlier implementations used ESP RainMaker primarily

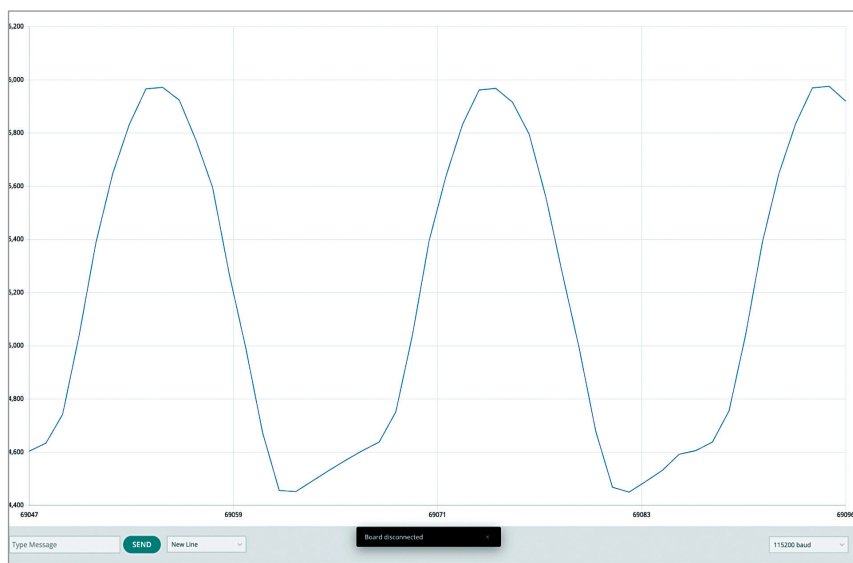


Fig. 5: Sine waveform observed on the serial monitor while rotating the potentiometer

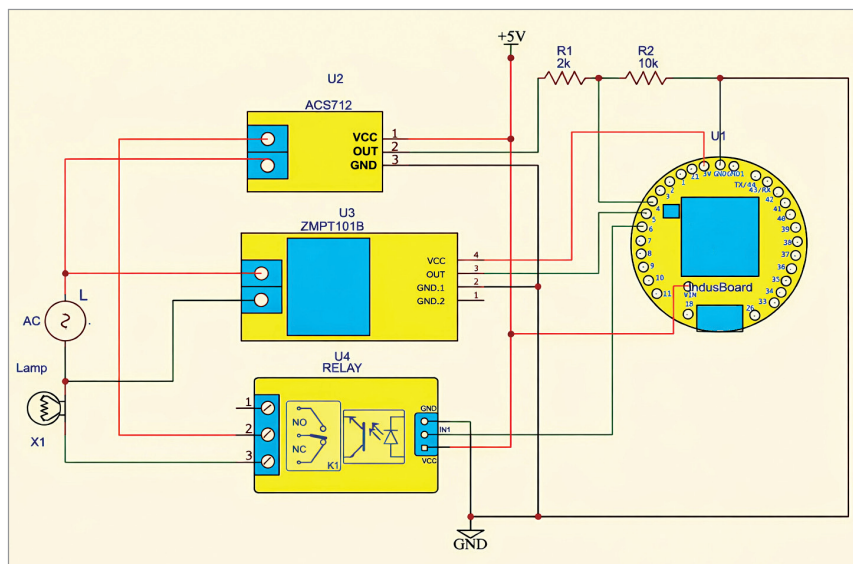


Fig. 6: Final circuit diagram